

DISK-STACKING AND FIBONACCI SPIRALS

A NEW OLD PARADIGM FOR MORPHOGENESIS

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An open problem

Most mathematicians know that Fibonacci spirals can be found in plants. Fewer know this topic offers rich open mathematical questions *and* generates hypotheses testable by modern molecular biology.

An old observation



Find the Fibonacci number!

Cylindrical lattices have two parameters & display parastichy numbers



Observational Data & Results

The reference dataset is taken from the MOSI Turing's Sunflower's project collection [cite: 210].

- There was a strong but not complete preponderance of Fibonacci counts [cite: 212].
- Excluding Fibonacci counts, the next commonest parastichy count was one less than a Fibonacci number (e.g., 33, 54, or 88) [cite: 213].
- This was followed by Lucas numbers (29, 47, or 76) and double-Fibonacci numbers (42, 68) [cite: 215].

[Figure 7 & 9:
Parastichy Fre-
quency Histograms]

Slow change rates support Fibonacci transitions [cite: 219]. However, rapid transitions promote near-Fibonacci counts, while noise promotes columnar structures [cite: 380]. Furthermore, double-Fibonacci and Lucas numbers emerge at large scales near the critical r' and are promoted by noise [cite: 433].

This work adds

The dynamics of slowly changing stacked-disk models naturally generate well-ordered lattice-like patterns with Fibonacci structure, providing a rationale for Turing's 'hypothesis of geometrical phyllotaxis' [cite: 455].

For faster rates of disk change, different parastichy counts emerge in ways highly compatible with observed data [cite: 456]. Ultimately, there is 'no gene for the number 55'; the observed structural motifs of phyllotaxis can only be understood in the light of a mathematical understanding of pattern dynamics [cite: 500].

Key Finding: The models account for a predominance of Fibonacci counts, while also explaining a smaller but detectable frequency of Lucas numbers, double Fibonacci numbers, and pairs of roughly equal but non-Fibonacci counts in a 'columnar' structure [cite: 5].

Disk-Stacking Methodology

Disk-stacking models were first introduced by Schwendener in the 1870s as an exploration of organ placement in plants [cite: 126]. The deterministic model is entirely defined by the function r_i , which gives the radius of the i th disk based on the height of the most recently placed disk [cite: 132, 133].

- **Control Parameter:** The slope r' is used as the control parameter, corresponding to the speed at which the disk size changes [cite: 134].
- **Stochasticity:** Noise is introduced by multiplying the radius by a scaling factor drawn from a uniform distribution of $[1 - \sigma, 1 + \sigma]$ [cite: 135].
- **Transitions:** Parastichy numbers in stacked-disk models change at γ and A dislocations [cite: 163]. Each jump corresponds to a γ dislocation of the parastichy lines, while A dislocations correspond to a decrease in the count [cite: 172, 173].

TL;DR

Disk stacking models are mathematically intriguing, computationally tractable, and have application to an old but still open biological problem: why do plants have Fibonacci numbers?